

Devika Mukherjee

Human Computer Interactions Researcher | Seeking PhD Opportunities

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HCI RESEARCH INTERESTS

Virtual and Mixed Reality (VR/XR), Haptics, Human-AI Interaction, Wearable Technologies, HCI in Healthcare Technologies

EDUCATION

09/2024 – 09/2025 **MSc Human-Computer Interaction, University of Birmingham**

09/2020 – 09/2024 **Bachelor of Engineering in Biotechnology, Amity University**

PUBLICATIONS

Mukherjee, D., Abdulkarim, D., Giunchi, D., Di Luca, M., Ofek, E. (2026).

“Belt and Whistles: Adding Lower Body Collision Awareness for MR Experiences.”

Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI 2026), Barcelona, Spain. <https://dl.acm.org/doi/10.1145/3772318.3793143>

RESEARCH EXPERIENCE

Research Associate (Grant writing and Industry Collaboration for Haptics).

ATOUGH Dec 25 – May 26

- Contributed to large-scale research funding acquisition through grant writing, industry partnership development (letters of support), and project coordination.

Research Associate (Haptic belt for VR)

VR Lab, University of Birmingham, June 25 – December 25

In Collaboration with : Prof. Eyal Ofek, Prof. Massimiliano Di Luca, Prof. Daniele Guinchi, Dr. Diar Abdulkarim

- Designed and conducted controlled VR experiments investigating embodiment, multisensory feedback, and user performance (n = 35+).
- Developed immersive experimental environments in Unity for behavioural studies in human–environment interaction.
- Collected and analysed behavioural, performance, and self-report data using Python.
- Evaluated the impact of wearable haptic feedback on perception, task performance, and user experience
- Research Output: First-author publication at CHI 2026.**

Unity Developer, Human-Robot Interaction

Extreme Robotics Lab, University of Birmingham December 25

- Designed and developed a high quality simulation** for battery dissemination by KUKA KR-16 robot for demonstration and integration with ML algorithm to control a KUKA KR-16 in real life.

Associate Event Coordinator (volunteer)

BhamXR, September 25-ongoing

- Coordinated and promoted **immersive tech showcases featuring XR, Robotics, Haptics across various human-centred technology domains.**
- Collaborated with entrepreneurs, educators and researchers** to organise **learning sessions and community feedback events** strengthening user-focused evaluation of emerging technologies.

Cytogenetics Laboratory & Data Analysis Intern

MedGenome Labs, January 24 – April 24

- Reduced laboratory labour by 15%.** Analysed cytogenetics data from laboratories for accurate diagnosis and aided in **reporting documentation.**
- Analysed Down’s syndrome case data to identify key trends, **enabling a 30% reduction in time-to-diagnosis.**
- Published **peer-reviewed paper** (solo author): [Link to publication.](#)

PROJECTS

Hydrino: ML Solution for developing healthier habits

- Designed an IoT-based hydration tracking product using Arduino Nano and a flutter based application. Using ML to track water bottle usage.

Better Diet: AI Nutrition Tracking App

- Conducted interviews and survey study (n=50+)** exploring dietary decision-making patterns.
- Developed **personas and behavioural journey maps analysing food selection motivations.** **Designed and prototyped interface interventions** to influence healthy consumption choices.

AI Bias in News Recommendation Systems

- Conducted **systematic literature review** and critical analysis of algorithmic bias in AI-driven news platforms; proposed **design guidelines for transparency, fairness auditing, and user-centred bias mitigation in intelligent interactive systems**.

Systematic Literature Review: Usability of Glucometers in T2DM Self-Management

- Synthesised findings across studies to **evaluate procedural usability, result interpretation, and integration with mobile health (mHealth) systems**.
- Identified key design factors (accessibility, error reduction, interface clarity) influencing user adherence and proposed improvements for **inclusive healthcare technologies**

Data Visualisation & Quantitative statistical UX Analysis

[GitHub](#)

- Investigated the relationship between in-game performance (score) and user experience (UX) metrics in “The Robot Game” using R. **Applied (Spearman, Wilcoxon) to explore correlations between player performance and usability ratings**.
- **Produced visualisations and insights** that informed interface recommendations and player feedback mechanisms.

REFERENCES

1. Prof. Eyal Ofek

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2. Prof. Massimiliano Di Luca

School of Computer Science, University of Birmingham, Edgbaston, Birmingham B15 2TT, United Kingdom

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